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Optical device

Abstract

An optical device according to an embodiment includes a lens mirror array and a holder. The lens mirror array is opposed to a light emitting section. The lens mirror array includes a plurality of optical elements. The optical elements include incident-side lens surfaces, emission-side lens surfaces that emit incident light, and reflection surfaces that reflect incident light. A cross section orthogonal to a first direction of the lens mirror array has a constricted shape. The holder includes a slit having width smaller than width of both the ends in the cross section of the lens mirror array and larger than width of the center in the cross section and extending in the first direction. The holder holds the lens mirror array in a state in which a constricted portion in the center in the cross section of the lens mirror array is disposed in the slit.

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Background/Summary

FIELD

(1) Embodiments described herein relate generally to an optical device including a lens mirror array, and an image forming apparatus containing the optical device.

BACKGROUND

(2) A printer or a multifunction peripheral includes, for example, as an exposure device for forming an electrostatic latent image on a photoconductive drum, a solid-state head including a light emitting element such as an LED. The solid-state head includes, for example, a substrate on which a plurality of light emitting elements are arrayed and mounted, a SELFOC lens array (SLA) disposed to be opposed to the plurality of light emitting elements, and a holder that positions and holds the substrate and the SLA. The SLA has structure in which a plurality of micro rod lenses are arrayed and embedded in resin. The solid-state head has approximately the same length as the length of the photoconductive drum.

(3) Since the SLA is made of long resin, the SLA is easily bent by an external force. If the SLA is bent, defocus occurs. Therefore, the rigidity of the holder that holds the SLA needs to be set as high as possible. Since the SLA has a short focal length and a small depth of field, the solid-state head needs to be disposed close to the photoconductive drum. Therefore, the solid-state head is desirably made as thin as possible. However, if the size of the holder is increased or a reinforcement member is attached to the holder to increase the rigidity of the holder, the width of the holder also increases to make it difficult to dispose the solid-state head close to the photoconductive drum.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram illustrating an example of an image forming apparatus including a solid-state head;
(2) FIG. **2** is a schematic diagram illustrating an example of the solid-state head;
(3) FIG. **3** is a perspective view illustrating a lens mirror array;
(4) FIG. **4** is a perspective view of the lens mirror array viewed from another angle;
(5) FIG. **5** is a perspective view illustrating a first example of a holder; and
(6) FIG. **6** is a perspective view illustrating a second example of the holder.

DETAILED DESCRIPTION

(7) An optical device according to an embodiment includes a lens mirror array and a holder. The lens mirror array is opposed to a light emitting section extending in a first direction. The lens mirror array includes a plurality of optical elements side by side in the first direction. The optical elements include incident-side lens surfaces on which light emitted from the light emitting section is made incident, emission-side lens surfaces that emit the incident light, and reflection surfaces that reflect the light made incident from the incident-side lens surfaces toward the emission-side lens surfaces. A cross section orthogonal to the first direction of the lens mirror array has a constricted shape, a center of which is narrower than both ends thereof that are along an optical path on which the light emitted from the light emitting section passes. The holder includes a slit having width smaller than width of both the ends in the cross section of the lens mirror array and larger than width of the center in the cross section and extending in the first direction. The holder holds the lens mirror array in a state in which constricted portion in the center in the cross section of the lens mirror array is disposed in the slit.

(8) The embodiment is explained below with reference to the drawings.

(9) An image forming apparatus **100** illustrated in FIG. **1** is a so-called multifunction peripheral including, for example, a print function, a copy function, and a scan function. The image forming apparatus **100** includes a housing **2**. A transparent original table glass **3**, on which an original is set, is present on the upper surface of the housing **2**. The original table glass **3** and a reading glass **5** disposed in parallel to the original table glass **3** are present on the upper surface of the housing **2**. The original table glass **3** and the reading glass **5** are disposed side by side in the left-right direction in FIG. **1** (a sub-scanning direction). An auto document feeder (ADF) **4** is present on the original table glass **3**. The ADF **4** is capable of opening and closing the original table glass **3**. The ADF **4** functions as an original cover for pressing the original placed on the original table glass **3** and has a function of feeding the original through the reading glass **5**.

(10) An original reading device **10** is present in the housing **2** under the original table glass **3**. The original reading device **10** is an example of the optical device described in the claims of the present application. The original reading device **10** is movable in the sub-scanning direction along the original table glass **3** by a not-illustrated driving mechanism and can be fixed under the reading glass **5** (in a position illustrated in FIG. **1**). The original reading device **10** extends in a main scanning direction (the first direction) orthogonal to the paper surface of FIG. **1** and causes a not-illustrated image sensor to form an erected image of an original.

(11) The original reading device **10** includes a lens mirror array **20** and a holder **54** having substantially the same structure as the structure of a solid-state head **504** explained below. Accordingly, the lens mirror array **20** and the holder **54** are explained in detail below in explanation of the solid-state head **504**. Explanation of the lens mirror array **20** and the holder **54** of the original reading device **10** is omitted.

(12) If an original is read, for example, the original reading device **10** is fixed under the reading glass **5** (a state illustrated in FIG. **1**), the original is fed by the ADF **4**, and the original is irradiated with illumination light via the reading glass **5**. The lens mirror array **20** guides reflected light reflected from the original and forms an image on the image sensor. A surface extending in the

main scanning direction on which the original receives the illumination light is a surface that reflects the light and is an example of the light emitting section described in the claims of the present application. The original reading device **10** photoelectrically converts the reflected light reflected from the original and received by the image sensor and outputs the reflected light as an image signal.

(13) At this time, the original reading device **10** reads, line by line in the main scanning direction, an erected image of the original passing on the reading glass **5** according to the operation of the ADF **4**. If the original passes on the reading glass **5** in the sub-scanning direction, the original reading device **10** can acquire an image of the entire original (for a plurality of lines). Alternatively, if the original is set on the original table glass **3** and the original reading device **10** is moved in the sub-scanning direction along the original table glass **3**, similarly, the original reading device **10** can read, line by line in the main scanning direction, an erected image of the original formed on the image sensor via the lens mirror array **20** and acquire an image of the entire original.

(14) The image forming apparatus **100** includes an image forming section **30** substantially in the center in the housing **2**. The image forming section **30** includes a yellow unit **301**, a magenta unit **302**, a cyan unit **303**, and a black unit **304** in a traveling direction of an intermediate transfer belt **40**. Since the color units **301**, **302**, **303**, and **304** of the image forming section **30** have substantially the same structure, the black unit **304** is representatively explained herein. Detailed explanation about the other color units **301**, **302**, and **303** is omitted.

(15) As illustrated in FIG. **2**, the black unit **304** includes, for example, a photoconductive drum **314**, an electrifying charger **324**, a solid-state head **504**, a developing device **334**, a primary transfer roller **344**, a cleaner **354**, and a blade **364**. Solid-state heads **501**, **502**, **503**, and **504** of the color units **301**, **302**, **303**, and **304** are examples of the optical device described in the claims of the present application. The intermediate transfer belt **40** is wound around a plurality of rollers and extended endlessly and travels in the counterclockwise direction in FIG. **2**.

(16) The photoconductive drum **314** has a rotation axis extending in the main scanning direction. The photoconductive drum **314** rotates in a state in which the outer circumferential surface thereof is set in contact with the surface of the intermediate transfer belt **40**. The primary transfer roller **344** is present on the inner side of the intermediate transfer belt **40** opposed to the photoconductive drum **314**. The photoconductive drum **314** is rotated by a not-illustrated driving mechanism in an illustrated arrow direction (the clockwise direction) at the same peripheral speed as the peripheral speed of the intermediate transfer belt **40**.

(17) The electrifying charger **324** uniformly charges the surface of the photoconductive drum **314**. The solid-state head **504** irradiates the surface of the photoconductive drum **314** with exposure light based on an image signal for color-separated black and forms an electrostatic latent image based on the image signal for black on the surface of the photoconductive drum **314**. The developing device **334** supplies black toner to the electrostatic latent image formed on the surface of the photoconductive drum **314** and forms a black toner image on the surface of the photoconductive drum **314**.

(18) The primary transfer roller **344** transfers the black toner image formed on the surface of the photoconductive drum **314** to be superimposed on toner images of the other colors. The cleaner **354** and the blade **364** remove toner remaining on the surface of the photoconductive drum **314**. The color toner images transferred to be superimposed one another on the surface of the intermediate transfer belt **40** move according to the traveling of the intermediate transfer belt **40**.

(19) A transfer roller pair **37** for transferring, onto paper P, the color toner images transferred to be superimposed one another on the surface of the intermediate transfer belt **40** is present on a downstream side of the black unit **304** in the traveling direction of the intermediate transfer belt **40**. One transfer roller **371** is present on the inner side of the intermediate transfer belt **40**. The intermediate transfer belt **40** is supported by the one transfer roller **371**. The other transfer roller **372** is opposed to the one transfer roller **371** across the intermediate transfer belt **40**.

(20) Referring back to FIG. **1**, a paper feeding cassette **61** in which a plurality of pieces of paper P of a predetermined size are stacked and stored is present near the lower end in the housing **2** of the image forming apparatus **100**. The paper feeding cassette **61** can be, for example, drawn out and received from the front surface of the housing **2**. A pickup roller **62** that picks up a piece of paper P at the upper end in a stacking direction among the pieces of paper P stored in the paper feeding cassette **61** is present at an illustrated right upper end of the paper feeding cassette **61**. The pickup roller **62** rotates with the circumferential surface thereof set in contact with the paper P to pick up the pieces of paper P one by one.
(21) A paper discharge tray **63** is present in an upper part in the housing **2**. The paper discharge tray **63** is present between the original table glass **3** and the image forming section **30** and discharges the paper P, on which an image is formed, into the body of the image forming apparatus **100**. A conveying path **64** for conveying the paper P picked up from the paper feeding cassette **61** in the longitudinal direction toward the paper discharge tray **63** is present between the pickup roller **62** and the paper discharge tray **63**. The conveying path **64** extends through a nip of the transfer roller pair **37** and includes a plurality of conveying roller pairs **641** and a not-illustrated conveyance guide. A paper discharge roller pair **631** for discharging the paper P to the paper discharge tray **63** is present at the terminal end of the conveying path **64**. The paper discharge roller pair **631** is rotatable in both of forward and backward directions.
(22) A fixing roller pair **65** is present on the conveying path **64** on the downstream side of the transfer roller pair **37** (the illustrated upper side). The fixing roller pair **65** heats and pressurizes the paper P conveyed via the conveying path **64** and fixes, on the surface of the paper P, a toner image transferred onto the surface of the paper P.
(23) The image forming apparatus **100** includes a reverse conveying path **66** for reversing the paper P, on one surface of which an image is formed, and feeding the paper P into the nip of the transfer roller pair **37**. The reverse conveying path **66** includes a plurality of conveying roller pairs **661** that nip the paper P and rotates to convey the paper P and a not-illustrated conveyance guide. A gate **67** that switches a conveyance destination of the paper P between the conveying path **64** and the reverse conveying path **66** is present on an upstream side of the paper discharge roller pair **631**.
(24) If an image is formed on the paper P, the image forming apparatus **100** rotates the pickup roller **62** to pick up the paper P from the paper feeding cassette **61** and conveys, with the plurality of conveying roller pairs **641**, the paper P toward the paper discharge tray **63** via the conveying path **64**. At this time, the image forming apparatus **100** feeds the color toner images transferred and formed on the surface of the intermediate transfer belt **40** into the nip of the transfer roller pair **37** to be timed to coincide with conveyance timing of the paper p, gives a transfer voltage to the color toner images with the transfer roller pair **37**, and transfers the color toner images onto the surface of the paper P.
(25) The image forming apparatus **100** conveys the paper P, onto which the toner images are transferred through the fixing roller pair **65**, to heat and pressurize the paper P, melts the toner images and presses the toner images against the surface of the paper P, and fixes the toner images on the paper P. The image forming apparatus **100** discharges the paper P, on which an image is formed in this way, to the paper discharge tray **63** via the paper discharge roller pair **631**.
(26) At this time, if a duplex mode for forming an image on the rear surface of the paper P as well is selected, the image forming apparatus **100** switches the gate **67** to the reverse conveying path **66** at timing immediately before the trailing end in a discharging direction of the paper P being discharged toward the paper discharge tray **63** passes through the nip of the paper discharge roller pair **631**, reverses the paper discharge roller pair **631**, and switches back and conveys the paper P. Consequently, the image forming apparatus **100** directs the trailing end of the paper P to the reverse conveying path **66** and reverses the paper P to feed the paper P into the nip of the transfer roller pair **37**.
(27) The image forming apparatus **100** forms, on the surface of the intermediate transfer belt **40**,

toner images based on image data formed on the rear surface of the paper P, causes the intermediate transfer belt **40** holding color toner images to travel, and feeds the color toner images into the nip of the transfer roller pair **37**. Further, the image forming apparatus **100** transfers and fixes the toner images on the rear surface of the reversed paper P, and discharges the paper P to the paper discharge tray **63** via the paper discharge roller pair **631**.

(28) The image forming apparatus **100** includes a control section **70** that controls operations of the mechanisms explained above. The control section **70** includes a processor such as a CPU and a memory. The processor executes a program stored in the memory, whereby the control section **70** realizes various processing functions. The control section **70** controls the original reading device **10** to thereby acquire an image from an original. The control section **70** controls the image forming section **30** to thereby form an image on the surface of the paper P. For example, the control section **70** inputs image data read by the original reading device **10** to the image forming section **30**. The control section **70** controls operations of the pluralities of conveying roller pairs **641** and **661** to convey the paper P through the conveying path **64** and the reverse conveying path **66**.

(29) The solid-state head **504** of the black unit **304** is explained below with reference to FIGS. **2** to **5**. Detailed explanation of the solid-state heads **501**, **502**, and **503** of the other color units **301**, **302**, and **303** is omitted because the solid-state heads **501**, **502**, and **503** have the same structure as the structure of the solid-state head **504**.

(30) As illustrated in FIG. **2**, the solid-state head **504** is separated from and opposed to the photoconductive drum **314** below the photoconductive drum **314** in FIG. **2**. The solid-state head **504** includes the lens mirror array **20**, a light source unit **52**, and the holder **54**. The components **20**, **52**, and **54** of the solid-state head **504** extend in the main scanning direction orthogonal to the paper surface parallel to the rotation axis of the photoconductive drum **314** and have substantially the same length as the length of the photoconductive drum **314**.

(31) The holder **54** holds the lens mirror array **20**. The holder **54** and the lens mirror array **20** of the solid-state head **504** have the same structures as the structures of the holder **54** and the lens mirror array **20** of the original reading device **10** explained above. The lens mirror array **20** of the solid-state head **504** is attached in a direction vertically reversed from the direction of the lens mirror array **20** of the original reading device **10**. The holder **54** of the solid-state head **504** fixes the light source unit **52**. The holder **54** fixes the light source unit **52** and the lens mirror array **20** in a state in which the light source unit **52** and the lens mirror array **20** are alighted with each other.

(32) As illustrated in FIGS. **2** to **4**, the lens mirror array **20** has structure in which a plurality of transparent optical elements **21** having the same shape are disposed side by side in the main scanning direction and integrated. FIG. **2** enlarges and illustrates a cross section of the lens mirror array **20** taken along a surface orthogonal to the main scanning direction between two optical elements **21** adjacent to each other. In this embodiment, the lens mirror array **20** is formed by integrally molding transparent resin. The lens mirror array **20** may be formed by transparent glass.

(33) The optical elements **21** of the lens mirror array **20** guide diffused light diffused from an object point O to form an image at an image forming point F. One optical element **21** causes lights from a plurality of object points O arranged in the main scanning direction to form images on an image surface. For example, the one optical element **21** causes light from the object points O arranged in width twice or three times as large as a pitch in the main scanning direction of the optical element **21** to form images on the image surface. The optical element **21** reflects, on two reflection surfaces **23** and **24**, light made incident on an incident-side lens surface **22** and emits the light via an emission-side lens surface **25** to form an erected image of the object point O at the image forming point F.

(34) In the solid-state head **504**, the object point O is present on a light emission surface of a light emitting element **521** explained below of the light source unit **52** and the image forming point F is present on the surface of the photoconductive drum **314**. That is, the lens mirror array **20** of the solid-state head **504** guides lights emitted from a plurality of light emitting elements **521** disposed

side by side in the main scanning direction and forms an image on the surface of the photoconductive drum **314**. Since the lens mirror array **20** includes the plurality of optical elements **21** side by side in the main scanning direction, the lens mirror array **20** forms an elongated image in the main scanning direction. In contrast, for example, in the original reading device **10** illustrated in FIG. **1**, the object point O is present on an original surface and the image forming point F is present on a light receiving surface of the image sensor. That is, the lens mirror array **20** of the original reading device **10** guides light reflected on the original surface and forms an image on the light receiving surface of the image sensor.
(35) The optical element **21** includes, on the surface thereof, an incident-side lens surface **22**, an upstream-side reflection surface **23**, a downstream-side reflection surface **24**, and an emission-side lens surface **25**. The incident-side lens surface **22**, the downstream-side reflection surface **24**, and the emission-side lens surface **25** are curved surfaces convex to the outer side. The upstream-side reflection surface **23** is a flat surface. The upstream-side reflection surface **23** is an example of the first reflection surface described in the claims of the present application. The downstream-side reflection surface **24** is an example of the second reflection surface described in the claims of the present application.
(36) An imaginary boundary surface (the cross section illustrated in FIG. **2**) between two optical elements **21** adjacent to each other in the main scanning direction is a surface orthogonal to the main scanning direction and is a surface generally orthogonal to the surfaces **22**, **23**, **24**, and **25** explained above. The cross section is an example of the cross section described in the claims of the present application but may be a cross section of the optical element **21** taken along an imaginary surface orthogonal to the main scanning direction in any position in the main scanning direction. In the lens mirror array **20** in which the plurality of optical elements **21** are integrally connected in the main scanning direction, the surfaces **22**, **23**, **24**, and **25** of the optical elements **21** are respectively continuous surfaces connected in the main scanning direction.
(37) Light made incident on the incident-side lens surface **22** of the optical element **21** is diverging light. The incident-side lens surface **22** converges the diverging light and directs the diverging light toward the upstream-side reflection surface **23**. Light reflected on the upstream-side reflection surface **23** and the downstream-side reflection surface **24** converges once and changes to diffused light thereafter and is emitted via the emission-side lens surface **25**. The emission-side lens surface **25** converges and emits the light reflected on the downstream-side reflection surface **24**. An optical path of light passing the center in the main scanning direction of the optical element **21** and passing the cross section orthogonal to the main scanning direction is indicated by a broken line. An optical path of light transmitted through the optical element **21** is reflected twice and bent. The width in the cross section of the optical path of the light transmitted through the optical element **21** is smaller in the center than both ends that are along the optical path. Therefore, the lens mirror array **20** includes a constricted portion **201** narrow in the center that is along the optical path.
(38) The lens mirror array **20** of the solid-state head **504** needs to cause light emitted from the light emitting element **521** to form an image on the surface of the photoconductive drum **314** without deviation such that deviation and distortion do not occur in an electrostatic latent image formed on the surface of the photoconductive drum **314**. That is, in order to form a high-quality image in the image forming apparatus **100** in this embodiment, the lens mirror array **20** having extremely high dimension accuracy without deviation and distortion is necessary. Since the lens mirror array **20** easily bends, the holder **54** that holds the lens mirror array **20** desirably has high rigidity not to cause a bend in the lens mirror array **20**. The optical element **21** of the lens mirror **20** has a small focal length and a small depth of field. Therefore, the solid-state head **504** needs to be disposed close to the photoconductive drum **314**.
(39) As illustrated in FIG. **5**, the holder **54** integrally includes a top wall **541** and two sidewalls **542**, **542**. The top wall **541** and the two sidewalls **542**, **542** of the holder **54** can be formed by, for example, bending one piece of rectangular sheet metal in two parts along the main scanning

direction. Therefore, a cross section of the holder **54** taken along any plane orthogonal to the main scanning direction has a U shape. A ridge portion between the top wall **541** and the sidewall **542** has a gently curving shape. The top wall **541** is a long and flat plate shape extending in the main scanning direction and is opposed to the surface of the photoconductive drum **314**. Since the solid-state head **504** is disposed close to the surface of the photoconductive drum **314**, it is desirable to reduce the width in the sub-scanning direction of the top wall **541** of the holder **54** as much as possible. The holder **54** can be formed by machining sheet metal such as stainless steel or iron or can be formed by resin or the like.

(40) The top wall **541** includes, in the center in the sub-scanning direction, a slit **5411** with a fixed width extending in the main scanning direction. The slit **5411** pierces through the top wall **541**. Both ends in the longitudinal direction of the slit **5411** respectively terminate in positions separated from both end portions in the longitudinal direction of the top wall **541**. Besides, the holder **54** includes two end walls **543** disposed at both ends in the main scanning direction. The outer circumferential portions of the end walls **543**, the end portion of the top wall **541**, and the end portion of the sidewall **542** are joined by welding or the like. The holder **54** has higher bending strength and is less easily deformed than the lens mirror array **20**. In other words, the lens mirror array **20** has lower bending strength and more easily bends than the holder **54**.

(41) The width in the sub-scanning direction of the slit **5411** is slightly larger than the width in the sub-scanning direction of the constricted portion **201** in the cross section of the lens mirror array **20** illustrated in FIG. **2**. The constricted portion **201** is, in the cross section orthogonal to the main scanning direction of the lens mirror array **20**, a portion narrower in the center than both ends that are along an optical path on which light emitted from the light emitting element **521** passes the lens mirror array **20**. The constricted portion **201** of the lens mirror array **20** indicates all portions of the lens mirror array **20** that can be disposed in the slit **5411**.

(42) Portions at both the ends along the optical path in the cross section of the lens mirror array **20** are all portions that are wider than the slit **5411** and cannot be disposed in the slit **5411**. The portions indicate an emission-side portion **202** further on the emission-side lens surface **25** side than the constricted portion **201** and an incident-side portion **203** further on the incident-side lens surface **22** side than the constricted portion **201**. The emission-side portion **202** projecting to the outer side of the holder **54** via the slit **5411** is an example of the projecting portion described in the claims of the present application.

(43) The emission-side portion **202** of the lens mirror array **20** is located on the outer side of the holder **54** via the slit **5411**. The incident-side portion **203** of the lens mirror array **20** is located on the inner side of the holder **54** (the light source unit **52** side). The constricted portion **201** of the lens mirror array **20** is located in the slit **5411** and located on the inner side of the holder **54**. That is, the end portion on the emission-side portion **202** side of the constricted portion **201** of the lens mirror array **20** is located in the slit **5411**. In other words, a portion between the downstream-side reflection surface **24** and the emission-side lens surface **25** of the lens mirror array **20** is located in the slit **5411**. If the constricted portion **201** is disposed in the slit **5411** and the lens mirror array **20** is attached to the holder **54** in this way, it is possible to reduce the width of the slit **5411** and maintain high rigidity of the holder **54**.

(44) A widened portion **5412** where the width of the slit **5411** is increased is present at one end of the slit **5411**. The widened portion **5412** is a hole piercing through the top wall **541** of the holder **54**. The width in the sub-scanning direction of the widened portion **5412** is larger than the width in the sub-scanning direction of the incident-side portion **203** of the lens mirror array **20**. The widened portion **5412** only has to have a size and a shape for enabling the incident-side portion **203** of the lens mirror array **20** to be inserted through the widened portion **5412** and can be formed in any shape.

(45) For example, if the lens mirror array **20** illustrated in FIGS. **3** and **4** is attached to the holder **54** illustrated in FIG. **5**, the lens mirror array **20** is bent, the incident-side portion **203** of the lens

mirror array **20** is inserted into the widened portion **5412** of the holder **54** from the one end in the longitudinal direction, and the constricted portion **201** of the lens mirror array **20** is disposed in the slit **5411** while being slid. The length in the main scanning direction of the widened portion **5412** is length with which the lens mirror array **20** does not interfere with the top wall **541** of the holder **54** if the lens mirror array **20** is bent and the constricted portion **201** is inserted through and disposed in the slit **5411**.

(46) After the constricted portion **201** of the lens mirror array **20** is disposed in the slit **5411** of the holder **54**, the lens mirror array **20** is positioned with respect to the holder **54** and the lens mirror array **20** is fixed to the holder **54**. The lens mirror array **20** is fixed to the holder **54** using, for example, an adhesive S. The lens mirror array **20** brings a positioning surface **2021** of the emission-side portion **202** into surface-contact with an outer surface **5413** of the holder **54** to position the positioning surface **2021** with respect to the holder **54**.

(47) The outer surface **5413** of the top wall **541** of the holder **54** and the surface of the lens mirror array **20** are fixed by the adhesive S. In a state in which the lens mirror array **20** is positioned with respect to the holder **54**, the inner surface of one sidewall **542** of the holder **54** and the surface of the lens mirror array **20** are fixed by the adhesive S. Fixing parts by the adhesive S are a plurality of parts separated in the main scanning direction and are positions not interfering with the incident-side lens surface **22**, the upstream-side reflection surface **23**, the downstream-side reflection surface and the emission-side lens surface **25** of the lens mirror array **20**.

(48) If the positioning surface **2021** of the emission-side portion **202** of the lens mirror array **20** is brought into surface-contact with the outer surface **5413** of the holder **54** as explained above, a positioning surface **2031** of the incident-side portion **203** of the lens mirror array **20** comes into surface-contact with the inner surface of one sidewall **542** of the holder **54**. In other words, the holder **54** has a shape in which the positioning surface **2031** of the lens mirror array **20** is in surface-contact with the inner surface of one sidewall **542** of the holder **54** in a state in which the positioning surface **2021** of the lens mirror array **20** is in surface-contact with the outer surface **5413** of the holder **54**.

(49) As illustrated in FIG. **2**, the light source unit **52** of the solid-state head **504** includes a substrate **522** on which the plurality of light emitting elements **521** are mounted side by side in the main scanning direction. The light source unit **52** is an example of the light emitting section described in the claims of the present application. The light emitting elements **521** may be disposed in one row or a plurality of rows extending in the main scanning direction. A driving circuit **523** is mounted on the surface of the substrate **522** on which the light emitting elements **521** are mounted. A connector **524** for power feed is fixed on the opposite surface of the surface of the substrate **522** on which the light emitting elements **521** are mounted. The substrate **522** is fixed to the inner surface of the sidewall **542** of the holder **54** by the adhesive S in a state in which the substrate **522** is positioned with respect to the holder **54**.

(50) The positioning of the substrate **522** with respect to the holder **54** is implemented by incorporating the substrate **522** in the holder **54** in which the lens mirror array **20** is positioned and fixed, causing the light emitting elements **521** to emit lights, detecting, with a camera, an image formed on an image surface via the lens mirror array **20**, and disposing the substrate **522** in a position where deviation does not occur in the image. After the substrate **522** is positioned with respect to the holder **54** in this way, the holder **54** and the substrate **522** are fixed using the adhesive S in a state in which a positional relation between the substrate **522** and the holder **54** is maintained.

(51) The plurality of light emitting elements **521** emit lights based on image data (an image signal) for black obtained by color-separating image data acquired by the original reading device **10** or image data acquired via external equipment such as a not-illustrated personal computer. The plurality of light emitting elements **521** are, for example, LEDs or OLEDs that emit lights or extinguish lights based on image data.

(52) The lights emitted from the plurality of light emitting elements **521** are made incident on the lens mirror array **20**. The lens mirror array **20** reflects and condenses the lights emitted from the light emitting elements **521** and emits light. The light emitted from the lens mirror array **20** is condensed on the surface of the rotating photoconductive drum **314**. At this time, an electrostatic latent image is written line by line in the main scanning direction on the surface of the photoconductive drum **314** according to the rotation of the photoconductive drum **314**. If the photoconductive drum **314** rotates a fixed amount, an electrostatic latent image for black obtained by color separation corresponding to an entire image of an original is formed on the surface of the photoconductive drum **314**.
(53) The holder **54** may be a holder **56** in which an opening section **5414** connected to one end of the slit **5411** is provided in one end wall **543** as illustrated in FIG. **6** instead of the widened portion **5412** provided at one end of the slit **5411**. In this case, the one end of the slit **5411** provided in the top wall **541** needs to be extended to the end wall **543** in which the opening section **5414** is provided. The opening section **5414** of the end wall **543** has a size and a shape for enabling the incident-side portion **203** of the lens mirror array **20** to be inserted through the opening section **5414**. In order to keep high rigidity of the holder **56**, the opening section **5414** is desirably closed by not-illustrated sheet metal or the like after the lens mirror array **20** is fixed to the holder **56**.
(54) If the lens mirror array **20** is attached to the holder **56** illustrated in FIG. **6**, one end in the longitudinal direction of the lens mirror array **20** is inserted through the opening section **5414** of the holder **56** and the constricted portion **201** of the lens mirror array **20** is disposed in the slit **5411** of the top wall **541**. At this time, the lens mirror array **20** only has to be slid in the longitudinal direction and does not need to be bent. Positioning of the lens mirror array **20** with respect to the holder **56** and bonding and fixing of the lens mirror **20** to the holder **56** and positioning and bonding and fixing of the substrate **522** of the light source unit **52** only have to be implemented in the same manner as in the embodiment explained above.
(55) A bending test for a solid-state head explained below was implemented in order to check a relation between the rigidity of the holder **54** (**56**) and the width of the slit **5411**.
(56) First, three test specimens of the solid-state head were prepared. A first test specimen A is the solid-state head **504** illustrated in FIG. **2** in which the lens mirror array **20** illustrated in FIGS. **3** and **4** is fixed to the holder **54** illustrated in FIG. **5**. A second test specimen B is a solid-state head in which the lens mirror array **20** illustrated in FIGS. **3** and **4** is fixed to the holder **56** illustrated in FIG. **6**. A third test specimen C is, as a comparative example, a solid-state head in which the SELFOC lens array of the related art is fixed to a slit of a holder. The holders of the third test specimen C include relatively wide slits that hold the SELFOC lens array.
(57) The three test specimens A, B, and C were formed by pieces of sheet metal having width of the top wall of 10 mm, length of the top wall of 340 mm, and height of the sidewall of 12 mm and having the same thickness. The width of slits of the test specimens A and B was small width for allowing the constricted portion **201** of the lens mirror array **20** to pass and was set to 20% of the width of the top wall. In contrast, the width of the slit of the test specimen C was large width for allowing the SELFOC lens array to pass and was set to 40% of the width of the top wall. That is, the width of the slit of the test specimen C was set to a double of the width of the slits of the test specimens A and B.
(58) The prepared test specimens A, B, and C were disposed in a horizontal posture in which the outer surfaces of the top walls face upward, spacers were disposed on the upper surfaces of both ends of the top walls of the test specimens A, B, and C, and reference surfaces parallel to the top walls of the test specimens A, B, and C were disposed above the top walls across these two spacers. In this state, the longitudinal direction center portions of the test specimens A, B, and C were pushed up toward the reference surfaces and amounts of the center portions of the test specimens A, B, and C approaching the reference surfaces were measured as bending amounts. Pressing forces for pushing up the centers of the test specimens A, B, and C gradually increased from 0 g.

Bending amounts of the test specimens A, B, and C at a point in time when a load of 500 g was further applied from a state in which the spacers at both the ends were in contact with the reference surface (a state in which the loads of the test specimens were supported) were measured. The bending amounts were evaluated by lengths in which the distances between the centers of the test specimens and the reference surfaces changed before and after of the bending.

(59) As a result of the measurement, the bending amount of the test specimen A was 100 μm, the bending amount of the test specimen B was 140 μm, and the bending amount of the test specimen C was 200 μm. From this result, it was found that the test specimen A has the highest rigidity and the test specimen C has the lowest rigidity. That is, it was found that the rigidity of the solid-state head can be increased by reducing the width of the slit provided in the top wall of the holder.

(60) As explained above, with the solid-state head **504** in this embodiment, since the lens mirror array **20** is fixed to the holder **54** (**56**) in a state in which the constricted portion **201** of the lens mirror array **20** is disposed in the slit **5411** of the holder **56** (**56**), it is possible to reduce the width of the slit **5411** as much as possible and maintain high rigidity of the holder **54** and increase the rigidity of the solid-state head **504**. It is possible to prevent a deficiency in which a bend occurs in the lens mirror array **20**.

(61) In the holder **54** of the solid-state head **504** in this embodiment, since the ridge portion between the top wall **541** and the sidewall **542** is bent, the width in the sub-scanning direction of the flat outer surface **5413** of the top wall **541** in surface-contact with the positioning surface **2021** of the lens mirror array **20** is smaller than the width between the outer surfaces of the two sidewalls **542** (that is, the width of the holder **54**). By providing the slit **5411** in the top wall **541**, the flat outer surface **5413** of the top wall **541** that can come into surface-contact with the positioning surface **2021** is further narrowed. Therefore, as in this embodiment, it is effective to reduce the width of the slit **5411** as much as possible. It is possible to, without increasing the width of the holder **54**, sufficiently secure the width of the flat outer surface **5413** of the top wall **541** that comes into surface-contact with the positioning surface of the lens mirror array **20**.

(62) According to this embodiment, it is possible to, without increasing the size of the holder **54** (**56**) and without separately pasting a reinforcement member, increase the rigidity of the holder **54** (**56**) by reducing the width of the slit **5411**. It is possible to form the solid-state head **504** in a compact size. Therefore, according to this embodiment, it is possible to dispose the solid-state head **504** having high rigidity near the photoconductive drum **314**.

(63) According to this embodiment, if the lens mirror array **20** is attached to the holder **54** (**56**), the constricted portion **201** of the lens mirror array **20** comes into slide-contact with the slit **5411** of the holder **54** (**56**). However, since the constricted portion **201** between the downstream-side reflection surface **24** and the emission-side lens surface **25** of the lens mirror array **20** is disposed in the slit **5411**, it is possible to prevent a change in an optical characteristic due to friction.

(64) In the solid-state head **504** in this embodiment, if the emission-side lens surface **25** disposed on the outside of the holder **54** is stained, the emission-side lens surface **25** is wiped by a brush or cloth. At this time, it is conceivable that an external force is likely to be applied to a projecting portion projecting from the slit **5411** of the holder **54** to damage the lens mirror array **20**. Accordingly, as in this embodiment, it is effective to dispose, in the slit **5411**, the constricted portion **201** between the downstream-side reflection surface **24** and the emission-side lens surface **25** of the lens mirror array **20** and reduce the size of a portion projecting to the outside of the holder **54** as much as possible.

(65) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of invention. Indeed, the novel apparatus and methods described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the apparatus and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims

and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. An optical device, comprising: a lens mirror array opposed to a light emitting section extending in a first direction, the lens mirror array including a plurality of optical elements side by side in the first direction, the plurality of optical elements respectively including incident-side lens surfaces on which light emitted from the light emitting section is made incident, emission-side lens surfaces that emit the incident light, and reflection surfaces that reflect the light made incident from the incident-side lens surfaces toward the emission-side lens surfaces, a cross section orthogonal to the first direction of the lens mirror array having a constricted shape, a center of which is narrower than both ends thereof that are along an optical path on which the light emitted from the light emitting section passes; and a holder including a slit having a width smaller than a width of both ends in the cross section of the lens mirror array and larger than a width of a center in the cross section and extending in the first direction, the holder holding the lens mirror array in a state in which a constricted portion in the center in the cross section of the lens mirror array is disposed in the slit.
2. The optical device according to claim 1, wherein the light emitting section comprises a plurality of light emitting elements and a substrate on which the plurality of light emitting elements are mounted side by side in the first direction.
3. The optical device according to claim 2, wherein the holder holds the substrate in a state in which the plurality of light emitting elements are positioned with respect to the lens mirror array.
4. The optical device according to claim 3, wherein width of the optical path passing the cross section of the lens mirror array is smaller in the center than at both the ends.
5. The optical device according to claim 3, wherein the slit of the holder includes a widened portion having a size for enabling a portion further on the incident-side lens surface side than the constricted portion of the lens mirror array to be inserted through the widened section.
6. The optical device according to claim 3, wherein the holder includes, at an end portion in the first direction, an opening section having a size for enabling a portion further on the incident-side lens surface side than the constricted portion of the lens mirror array to be inserted through the opening section.
7. The optical device according to claim 3, wherein the holder holds the lens mirror array in a state in which a portion between the reflection surfaces and the emission-side lens surfaces of the plurality of optical elements of the lens mirror array is disposed in the slit.
8. The optical device according to claim 7, wherein the lens mirror array includes a projecting portion projecting to an opposite side of the light emitting section via the slit, the projecting portion including a positioning surface opposed to an outer surface of the holder, and the holder brings the positioning surface of the lens mirror array into surface-contact with the outer surface to position the lens mirror array.
9. The optical device according to claim 3, wherein the lens mirror array has a bending strength lower than a bending strength of the holder.
10. The optical device according to claim 3, wherein the reflection surfaces include a first reflection surface that reflects the light made incident from the incident-side lens surfaces and a second reflection surface that reflects the light reflected by the first reflection surface toward the emission-side lens surfaces, and the lens mirror array forms an erected image at an image forming point of the lens mirror array.
11. An image forming apparatus, comprising: a photoconductive drum; a supply of toner; and an optical device, comprising: a lens mirror array opposed to a light emitting section extending in a first direction, the lens mirror array including a plurality of optical elements side by side in the first direction, the plurality of optical elements respectively including incident-side lens surfaces on

which light emitted from the light emitting section is made incident, emission-side lens surfaces that emit the incident light, and reflection surfaces that reflect the light made incident from the incident-side lens surfaces toward the emission-side lens surfaces, a cross section orthogonal to the first direction of the lens mirror array having a constricted shape, a center of which is narrower than both ends thereof that are along an optical path on which the light emitted from the light emitting section passes; and a holder including a slit having a width smaller than a width of both ends in the cross section of the lens mirror array and larger than a width of a center in the cross section and extending in the first direction, the holder holding the lens mirror array in a state in which a constricted portion in the center in the cross section of the lens mirror array is disposed in the slit.

12. The image forming apparatus according to claim 11, wherein the light emitting section comprises a plurality of light emitting elements and a substrate on which the plurality of light emitting elements are mounted side by side in the first direction.

13. The image forming apparatus according to claim 12, wherein the holder holds the substrate in a state in which the plurality of light emitting elements are positioned with respect to the lens mirror array.

14. The image forming apparatus according to claim 13, wherein width of the optical path passing the cross section of the lens mirror array is smaller in the center than at both the ends.

15. The image forming apparatus according to claim 13, wherein the slit of the holder includes a widened portion having a size for enabling a portion further on the incident-side lens surface side than the constricted portion of the lens mirror array to be inserted through the widened section.

16. The image forming apparatus according to claim 13, wherein the holder includes, at an end portion in the first direction, an opening section having a size for enabling a portion further on the incident-side lens surface side than the constricted portion of the lens mirror array to be inserted through the opening section.

17. The image forming apparatus according to claim 13, wherein the holder holds the lens mirror array in a state in which a portion between the reflection surfaces and the emission-side lens surfaces of the plurality of optical elements of the lens mirror array is disposed in the slit.

18. The image forming apparatus according to claim 17, wherein the lens mirror array includes a projecting portion projecting to an opposite side of the light emitting section via the slit, the projecting portion including a positioning surface opposed to an outer surface of the holder, and the holder brings the positioning surface of the lens mirror array into surface-contact with the outer surface to position the lens mirror array.

19. The image forming apparatus according to claim 13, wherein the lens mirror array has a bending strength lower than a bending strength of the holder.

20. The image forming apparatus according to claim 13, wherein the reflection surfaces include a first reflection surface that reflects the light made incident from the incident-side lens surfaces and a second reflection surface that reflects the light reflected by the first reflection surface toward the emission-side lens surfaces, and the lens mirror array forms an erected image at an image forming point of the lens mirror array.